An app for detecting ‘Dark Vessels’ in EU Waters

Team Gamma

What are dark vessels?

- "Dark Vessels" are ships that intentionally switch off their Automatic Identification System (AIS) transponders to become digitally invisible, typically to hide illegal activities like unauthorized fishing, smuggling, or bypassing international sanctions.

General idea

- Build a monitoring dashboard that "unmasks" hidden maritime activity by cross-referencing what satellites see with what ships report.
- The ocean depends on a "handshake" system (AIS). When a captain turns it off, they create a "data blind spot" where environmental and labor laws no longer apply.
- We want to create an app that watches over European waters and flags discrepancies between physical presence and digital signals

Possible datasets

- **AIS Hub (API):** Provides real-time GPS, name, and flag of every ship currently "behaving" and broadcasting.
- **Sentinel-1 SAR (Copernicus Hub):** Provides radar imagery that detects the metallic hull of a ship.
- **EMSA (European Maritime Safety Agency) Port Data:** To verify if a "detected but silent" ship was recently cleared to leave a European port.

Possible benefits for the user

- If an oil spill is detected via satellite, our app can look back in time to see which vessels were in the area, even if they turned their trackers off.
- Identifying potential illegally sailing ships that stay offshore and invisible to avoid port inspections.

Possible extra features

- Use a simple algorithm to flag "Impossible Travel"; if a ship’s AIS says it’s in the Mediterranean, but the satellite sees a matching hull shape in the Baltic Sea.

Tech stack

- Frontend: React.js
- Backend: GO
- Database (if needed): PostgreSQL
- Hosting: Firebase